

ORIGINAL ARTICLE

Positive deviance in child feeding and nutrition practices among mothers in Northern Benin

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ABSTRACT

Background and aim: Positive Deviance (PD) is an approach based on the principle that solutions to a community's problems already exist locally. These solutions are considered to be healthy, affordable, culturally acceptable, and sustainable solutions to malnutrition. The goal of this study was to identify child feeding and nutrition practices used by PD mothers.

Methods: A 6-stage Positive Deviance Inquiry (PDI) was conducted through semi-structured interviews (24) and direct observations (24) with PD, Non-PD and negative deviant (ND) mothers. They were selected on the basis of specific criteria from the *Baatonu*, *Gando* and *Fulani* ethnic groups in the communes of *Banikoara* and *Nikki* in northern Benin. The interview recordings were transcribed and a content analysis was carried out using MAXQDA software.

Results: Four categories of practices were identified, notably feeding, care, hygiene and health. In terms of feeding practices, PD mothers use lactation-enhancing methods and "Gentle" breastfeeding cessation method to ensure adequate child feeding. The following food recipes derived from combinations of nutritionally interesting local resources were often served by PD mothers to their children: *Idouwarê*, *Dambu*, *Dibou kpé sari*, *Kobsa* and cow's milk sauce. Concerning care and hygiene practices, PD mothers had a more affectionate mother-child relationship, covered their food and drinking water and kept their yard and kitchen clean. In terms of health practices, PD mothers often referred their children to the hospital after self-medication when they had diarrhea.



Received: 20 January 2026 | Accepted: 25 May 2026

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Conclusions: These PD practices could be promoted through nutrition education sessions using communication tools for behavior change.

Key words: positive deviant, child, malnutrition, feeding practices, local resources

Introduction

Malnutrition is the greatest risk factor for mortality and morbidity among young children in developing countries. According to Black et al. (1), it is responsible for 45% of annual deaths of children under five. Despite the efforts of the Beninese government to reduce the prevalence of stunting by 3.2% from 2009 to 2019, malnutrition remains a major public health problem among children under five years. In northern Benin, particularly in *Borgou* and *Alibori* departments, more than 32% of children are stunted, 5% are wasted and more than 18% are underweighted (2,3).

Among the immediate causes of this situation, inappropriate breastfeeding and complementary feeding practices are highly prevalent in rural communities (4,5). Mothers are the primary caregivers responsible for child feeding. Their nutrition and feeding practices play a crucial role in addressing child malnutrition (6,7).

The innovative interventions such as the Positive Deviance (PD) approach that encourages positive and sustainable change in maternal behaviors regarding child feeding and nutrition can be very effective in helping to reduce child malnutrition. Positive Deviance is an approach based on the belief that every community has certain individuals ("Positive Deviants") who engage in unconventional practices and behaviors that enable them to find better ways to prevent malnutrition than their neighbors, even though they have the same resources and face the same risks. This approach is grounded in the principle of identifying sustainable and community-based solutions. In addition, given their strong societal embeddedness, the community-based solutions adoption are sustainable and affordable (6,8).

PD approach is used as an alternative to other nutritional approaches to improve the nutritional status

of children under five years in many countries including Benin (9,10). This approach has been shown to be effective in improving child nutrition and maintaining beneficial behaviors for children in the long term (11). Thus, in their study on the effectiveness of a PD approach in Ethiopia, Gizaw et al. (12) proved an improvement in mothers' nutritional knowledge, attitude and self-efficacy, as well as a reduction in children's underweight.

One of the keys to success in PD interventions is the availability of local food resources (6). In Benin, local resources are enormous but underused in terms of food agrobiodiversity (13–15). The availability of these resources is an opportunity, and their promotion through appropriate practices will improve the quality of children's diets to ensure their food and nutritional security (4,16).

This study aims to identify, at the community level, positive deviant practices regarding child feeding and nutrition among mothers of children aged 6 to 59 months from different ethnic groups in the communes of *Nikki* and *Banikoara*.

Material and Methods

Analytical framework

The Positive Deviance inquiry (PDI) approach used in this study was based on the methodology developed by the Child Survival Collaborations and Resources Group (6). Thus, it was carried out in six steps (see Figure 1).

Study setting

The study was conducted in the communes of *Nikki* and *Banikoara* during January to February 2023

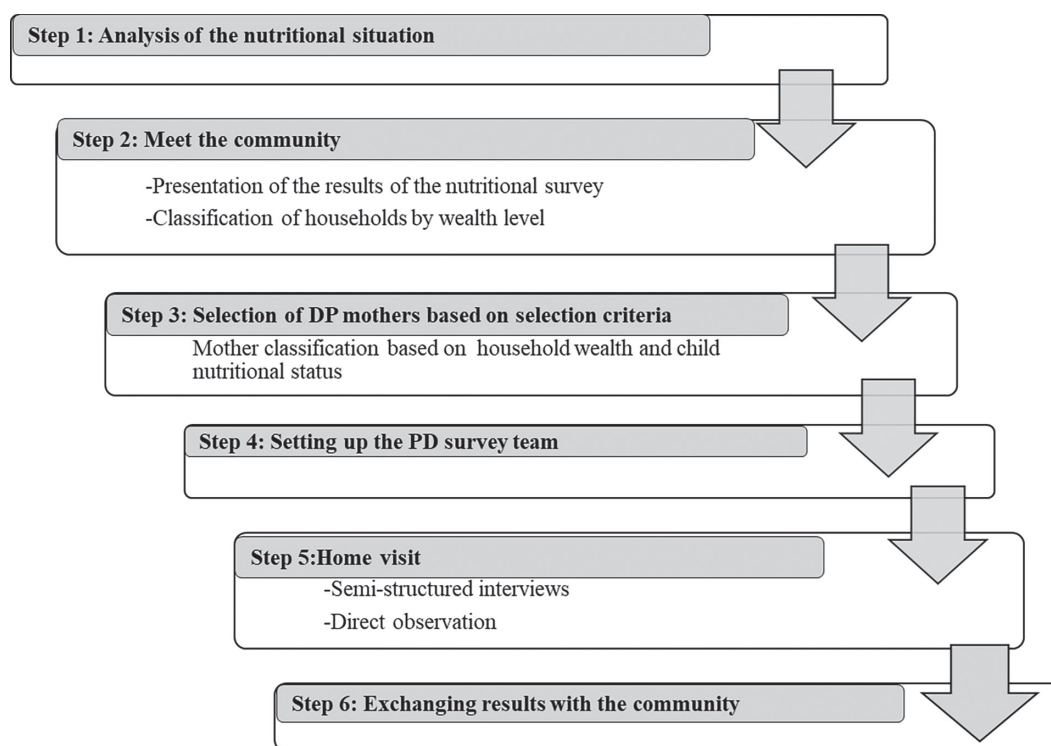


Figure 1. Process of positive deviance inquiry.

among three ethnic groups, *Baatonu* (village of *Guinrou*), *Gando* (village of *Alafiarou*) and *Fulani* (village of *Nimbéré-Peulh*).

Sample

Semi-structured interviews and direct observations were conducted with a total of 24 mothers, eight mothers in each of the three villages. Participants were selected in agreement with community members according to three criteria: (1) nutritional status of the child, (2) level of household wealth and (3) being recognized within the community for its uncommon practices.

Data collection

This PDI was carried out in six steps defined in the theoretical framework previously presented.

Analysis of the nutritional situation

The nutritional survey was conducted among children aged 6 to 59 months in the three villages to

identify those who are stunted, wasted and underweight. Age and anthropometric data (weight and height) were collected from all children in the village. They were classified as stunted, wasted and underweight using the height-for-age, weight-for-height and weight-for-age indexes, respectively. In the village of *Guinrou*, out of 82 children surveyed, 41 were malnourished, including 24 stunted, five wasted and 12 underweight. In *Alafiarou*, 101 out of 115 children surveyed, were malnourished, including 62 stunted, eight wasted and 31 underweight. Among 48 children surveyed, 45 were malnourished, including 20 stunted, seven wasted and 18 underweight in the village of *Nimbéré-Peulh*.

Meeting with the community

The meeting was organized with the leaders of the women's groups, the community health volunteers and the village chief. The objective was to share with them the results of the nutritional status of their children, to define household wealth and to identify positive deviant

mothers. Then, the nutritional status of children was presented to participants in terms of the percentage of malnourished children using a pie chart to help them understand the severity of malnutrition in their community.

The characteristics of wealthy households defined by the participants were as follows:

- Type of building materials of the house: the house of the rich is not built of banco bricks but of cement bricks.
- Motorcycle ownership: The head of a wealthy household owns a motorcycle.
- Possession of cattle (beef): The head of a wealthy household owns more than five beefs.
- Land: The head of a wealthy household owns at least five hectares of arable land.
- Owning a mill: The wealthy household owns a small condiment mill.
- Quality of clothing materials: The heads of wealthy households always wear loincloths costing more than 1,333 francs CFA per meter.
- Children's schooling: The head of a wealthy household educates (school) their children.

All ethnic groups have the first four characteristics in common. The fifth characteristic is specific to *Gando*, while the seventh is specific to the *Baatonu*. The *Baatonu*'s and the *Fulani*'s share the last two characteristics. Using these characteristics, the participants themselves classified the surveyed households into two categories: poor and wealthy.

Selection of PD mothers

Based on the previous criteria, mothers were classified into four categories:

- Positive deviant mother: Mother from a poor household with a well-nourished child.
- Non-positive deviant (non-PD) mother: Mother from a poor household with a malnourished child.
- Negative deviant (ND) mother: Mother from a wealthy household with a malnourished child.
- Normal mother: Mother from a wealthy household with a well-nourished child.

Four mothers with good child feeding and nutrition practices (PD) and another four with inadequate child feeding and nutrition practices (non-PD, ND) were selected.

In each village, three PD mothers, two non-PD mothers and two ND mothers were randomly selected to participate in the study. In order to be sure that mother was really PD, the following criteria were checked:

- The mother must have at least two children.
- The mother should not have children suffering from severe malnutrition.
- The mother's household must not have any serious or atypical social or health problems.
- Her child must be in good health at the time of the visit.
- His child must not have a beggar's personality.
- Her child should not be a very large baby who has lost weight recently.
- Her child should not be a very small baby with a low birth weight but who is growing well today.
- Her child must not have any serious health problems.

To take into account the participatory aspect of the PDI approach, participants were also asked to identify by consensus in the community, one mother with good child feeding and nutrition practices to be part of the study.

Setting up the PDI team

The PDI team was composed by the investigator, the head of the women's group, and the translator. Before the beginning of the activity, the procedures for conducting home visits were explained to the head of the women's group.

Home visit

A home visit was organized in the household of each selected mother during which a semi-structured interview was conducted on their feeding practices (early initiation of breastfeeding, age of introduction

of solid foods, frequency of breastfeeding, introduction of herbal teas, foods consumed and their frequency, dietary diversity, age of cessation of breastfeeding and methods used, etc.), care practices (caregivers and their relationships with children), hygiene practices (children's personal hygiene, food hygiene and drinking water, etc.) and health practices (frequency of illnesses such as diarrhea and their treatment).

The home visits were conducted during the preparation and consumption of food in the household. Direct observations based on child and parent interaction and appearance, feeding practices, water quality, hygiene and sanitation have also been done. Direct observations were used to triangulate the information provided during the semi-structured interviews and to collect other complementary information (17). The duration of the semi-structured interviews varied between 45–90 minutes.

Sharing PDI results with the community

The good child feeding and nutrition practices identified during the survey were shared with the mothers. The discussion was based on the relevance of each PD child feeding and nutrition practice identified.

Data processing and analysis

The interviews were transcribed by an expert in the local language and enriched with field notes. The transcripts were then checked by a member of the team. After this verification, they were analyzed by coding using the content analysis approach with MAXQDA software. The first codes were generated using the main themes of the questionnaire. Subcodes were inferred as transcriptions were analyzed. The transcripts were coded by the third author and checked by the first author. These codes were then shared with the co-authors for validation. Memos were used to explain the importance of the codes in relation to the purpose of the study (18). Similar codes have been grouped together to form themes.

For dietary diversity data, foods were grouped into eight food groups to determine the child's dietary diversity score (19). The frequency of foods consumed

by children was categorized as follows: once a day (frequent); 2 to 6 times a week (often); at most once a week (scarce). Age of introduction of complementary foods, age of cessation of breastfeeding, dietary diversity score and daily consumption frequencies mean were calculated for comparison. Food recipes were classified as PD based on two criteria: (1) food recipes served only by PD mothers or (2) food recipes more frequently consumed than other mothers.

In order to facilitate interpretation, the data were divided into two groups: the group of PD mothers who are supposed to have good child feeding and nutrition practices and the group of non-PD mothers and ND mothers who have inadequate child feeding and nutrition practices.

Results

Socio economic and demographic characteristics of mothers

Table 1 shows that all mothers over 45 years old were PD. In addition, the majority of mothers having more than six children were PD. In terms of education, almost all mothers (95.83%) have no formal level of education. In both categories, more than half (13) of the mothers were engaged in income-generating activities.

Feeding practices

The feeding practices presented in Table 2 were related to breastfeeding, complementary feeding, food recipes and cessation of breastfeeding.

A. BREASTFEEDING

Across all ethnic groups and categories of mothers, colostrum was administered at birth by all mothers. Mothers reported breastfeeding their children on demand. In cases of breastfeeding difficulties, traditional recipes transmitted by grandmothers were used to facilitate or restore breast milk production. A *Baatonu PD mother* and all *Gando PD* mothers used lactation-enhancing methods such as the consumption

Table 1. Socio economic and demographic characteristics of mothers.

Characteristics	All mothers (n=24)	Positive deviant mothers (n=12)	Non-positive deviant/negative deviant mothers (n=12)
Mother's age			
15-29	14(58.33%)	4 (33.33%)	9(75%)
30-44	7 (29.17%)	4(33.33%)	3 (25%)
≥45	3(12.50%)	4(33.33%)	0(0%)
Number of children			
≤ 6	16(66.67%)	6(50%)	10(83.33%)
>6	8 (33.33%)	6(50%)	2 (16.66%)
Level of education			
Illiteracy	23(95.83%)	11(91.16%)	12
Primary	0(0%)	0(0%)	0(0%)
Secondary	1 (4.17%)	1(8.33%)	0(0%)
Income-generating activities			
Yes	13(54.16%)	7 (58.33%)	6(50%)
No	11 (45.83%)	5 (41.66%)	6(50%)

of porridge made from a decoction of *Euphorbia Balsamifera* leaves called *Tcholi* (*Gando*), or porridge made from millet or sorghum grits called *bita* (*Baatonu*).

B. COMPLEMENTARY FEEDING PRACTICES

General practices

Besides *Fulani* mothers, half of *Baatonu* and *Gando* PD mothers gave their children porridge enriched with legumes or prepared special sorghum porridge (light porridge) for children with cow's milk. Non-PD/ND mothers fed their children with family porridge, which is often very consistent. Except for *Baatonu* mothers, the average age of introduction of complementary foods is higher (8 ± 1.7 months and 6 ± 0 months compared to 5.5 ± 0.58 months and 5 ± 0 months) among PD mothers than among non-PD/ND mothers in *Gando* and *Fulani* ethnic groups. Only *Baatonu* PD mothers gave herbal teas to their children before this age of complementary food introduction, which is not the case for non-PD/ND mothers.

The quantity of meals served to children by PD mothers is greater than that served to the children of non-PD/ND mothers among the *Baatonus* and *Fulanis*, with the exception of the *Gandos*. Once the meal

was served, the children of PD mothers were actively fed at *Baatonus* and *Gandos*.

Buying cooked food outside the household was a practice that *Baatonu* and *Gando* PD mothers avoided. In *Fulani* settings, non-PD/ND mothers frequently fed their children with foods from outside the household or allowed them to eat with neighbors, which is not the case for PD mothers. In addition, *Baatonu* and *Gando* PD mothers used several methods (decoction of *Vernonia amygdalina*, eggs prepared with the decoction of *Daniela oliveira* leaves, etc.) to stimulate children's appetite.

The frequency of daily food consumption by children was also a practice that differed between the two categories of mothers in all ethnic groups. Children of PD mothers have a daily frequency of consumption up to six compared to other children with a maximum frequency of four. Among the *Baatonus* and *Fulanis*, children of PD mothers appeared to consume more food groups (>5) than other children (>3).

Food Recipes

○ Food recipes offered by PD mothers to their children

The food recipes served by PD mothers to their children were divided into two categories: The main

Table 2. Feeding practices of different categories of mothers in the three ethnic groups.

Categories	Mothers' practices in child feeding		
	Common	Positive deviant	Non-positive deviant/negative deviant
Feeding			
Breastfeeding	- Colostrum intake by children at birth^{BGF} - Breastfeeding on demand^{BGF} - No use of the lactation-enhancing method^F	- Use of lactation-enhancing Methods: - Mother consumption of millet porridge or sorghum to stimulate breast milk production^B - Use of <i>Euphorbia basalmifera</i> leaf decoction prepared with the porridge for lactation-enhancing^G	No use of lactation-enhancing methods^{BG}
Complementary feeding	The child drinks the porridge intended for the whole household^F	- The complementary foods were: - sorghum porridge + cow's milk or enriched porridge^B - special sorghum porridge for children (light porridge) or enriched porridge^G	- The complementary foods were: - Sorghum porridge in some cases, Nigerian yellow semolina^B - Sorghum porridge intended for the whole household^G
	Introduction of herbal tea before the age of complementary feeding^{GF}	- Introduction of herbal tea before the age of complementary feeding^B - Average age of complementary feeding was: 5.77±0.50 months^B 8±1.73 months^G 6±0 months^F	- No introduction of herbal teas before the age of complementary feeding^B - Average age of complementary feeding was: 6±0.82 months^B 5.5±0.58 months^G 5±0 months^F
	Passive feeding^F	Active feeding^{BG}	Passive feeding^{BG}
	No common practices^{BGF} -	- The amount of food consumed by the child at each meal compared to the reference bowl* is often: 0.69±0.24^B 0.50±0^G 0.50±0^F	- The amount of food consumed by the child at each meal compared to the reference bowl* is often: 0.50±0^B 0.63±0.25^G 0.25±0^F
	No common practices^{BGF}	- The child consumed the day before the survey: 4±0 food groups^B 3.75±0.5 food groups^G 4.75±0.5 food group^F	- The child consumed the day before the survey: 3±0.82 food groups^B 3.75±0.96 food groups^G 3.25±0.50 food groups^F
	No common practices^{BGF} -	- The daily consumption frequency of child was: 4±0 times^B 4.75±0.50 times^G 4.8±0.50 times^F	- The daily consumption frequency of child was: 3±0 times^B 3.6±0.58 times^G 3.8±1.26 times^F
	Child rarely eats food outside the household^{BG}	Child rarely eats outside the household^F	Child often eats out of the household^F

Table 2 (Continued)

Categories	Mothers' practices in child feeding		
	Common	Positive deviant	Non-positive deviant/negative deviant
	The mother did not use an appetite stimulation method^F	- Use of appetite stimulation methods: - dietary change - use of <i>Vernonia amygdalina</i> (<i>Tuan</i> in <i>Baatonu</i>) leaves^B - clean the child's mouth with charcoal using cotton^B - eat an egg prepared with a decoction of <i>Daniela oliveira</i> leaves (<i>Kalabi</i> in <i>Fulani</i>)^G - clean the child's mouth with the crushed leaves of <i>Arbus precatarius</i> (<i>Boduboduel</i> in <i>Gando</i>)^G	No use of appetite stimulation method^{BG}
	Rare street food purchase^F	- Rare purchase of street food^B - Frequent purchase of street food^G	- Frequent purchase of street food^B - Rare purchase of street food^G
Food recipes	- Main food recipe - The child frequently eats pasta (maize or sorghum) and often yam stew and spaghetti^B	- Main food recipe - Children often consume <i>Idouwaré</i> (<i>Gourmantché</i>): cowpea sorghum grits seasoned with peanut paste; <i>Dibou kpé sari</i> (<i>Baatonu</i>) or <i>Balidjalo</i> (<i>Gourmandjé</i>): sorghum grits seasoned with <i>Urena Lobata</i> (<i>Boubou djan mou</i> in <i>Baatonu</i>); <i>Dambu</i> (<i>Fulani</i>): maize couscous with cowpeas; <i>Dambu</i> (<i>Baatonu</i>): maize couscous seasoned with <i>Senna obtusifolia</i> leaves (<i>Kpankpanti</i> in <i>Baatonu</i>) and rarely prepared egg^B	- Main food recipe - The child often consumes <i>Akassa</i> (fermented maize dough) and <i>Wake</i> (rice + cowpea)^B
	The child frequently eats maize or sorghum paste and often rice and yam stew^G	Cowpeas, eggs and couscous are rarely prepared^G	The child frequently consumes pounded yam, spaghetti and often prepared yam. Recipes such as <i>Wassarwassa</i> (<i>couscous made from yam cossette flour</i>), <i>sotè</i> (<i>Baatonu</i>) (seasoned maize or sorghum pellets + cowpeas) are rarely eaten^G
	The child frequently eats maize or sorghum paste and often rice^F	<i>Dambu</i> (maize couscous) is rarely prepared^F	Spaghetti is rarely prepared^F
	- Side recipe - The child frequently consumed dry okra sauce and baobab leaf powder and often sesame sauce, guinea sorrel sauce and peanut sauce^B	- Side recipe - The child frequently consumed the <i>Ceratotheca sesamoides</i> Endl. sauce (<i>Kpewori</i> in <i>Baatonu</i>) and often cowpea and okra leaves sauce (<i>Kobsa</i> in <i>Baatonu</i>)^B	- Side recipe - The child often consumed tomato sauce and sauce made from okra leaves^B

Categories	Mothers' practices in child feeding		
	Common	Positive deviant	Non-positive deviant/negative deviant
	The child frequently consumed dry okra sauce, baobab powder, <i>Sesamum radiatum</i> L. (<i>dossiho</i> in <i>Gando</i>) but rarely sesame, (<i>gèrouwa</i> in <i>Gando</i>) sauce, squash sauce and eggplant (<i>Boro</i> in <i>Gando</i>) sauce^G	The child often consumed <i>kobsa</i> (<i>Baatonu</i>) sauce, sorrel sauce, peanut sauce, and cow's milk sauce^G	The child often consumed the tomato sauce^G
	The child often consumed the baobab leaf powder sauce^F	The child often consumed <i>kobsa</i> (<i>Baatonu</i>), Guinea sorrel sauce and peanut sauce^F	The child often consumed the tomato sauce^G
Cessation of breastfeeding		- The average age for cessation of breastfeeding: 26.66±3.21 months^B 23±3.37 months^G 19±4.08 months^F	- The average age for cessation of breastfeeding: 19.25±3.95 months^B 20.75±3.66 months^G 25.25±8.58 months^F
	Use of “non-gentle” methods for cessation of breastfeeding^F	“Gentle”^B cessation of breastfeeding (cheese + breast milk)^G	Abrupt cessation of breastfeeding^G

^B= *Baatonu*, ^G= *Gando*, ^F= *Fulani*, Several exponents represent several ethnic groups, ^{*}Reference bowl: 1/3 liter, ^B “Gentle” methods, as opposed to ‘non-gentle’ methods, enable mothers to do the cessation of breastfeeding without encountering significant difficulties, such as crying or a refusal to eat.

dish and side dish. Across all ethnic groups, the food recipes served as the main dish by PD mothers were mainly cowpeas; *Idourwarê* (*Baatonu*), a dish of cowpea, sorghum grits seasoned with peanut paste; *Dibou kpé sari* (*Baatonu*) based on sorghum grits seasoned with *Urena Lobata* (*Bouhou djan mou* in *Baatonu*) leaves, *Dambu* (*Baatonu*), a maize couscous seasoned with *Senna obtusifolia* (*Kpankpanti* in *Baatonu*) leaves. Prepared eggs were also served as a main dish. The side dish included *kobsa* (*Baatonu*) sauce, made from dried okra leaves with cowpeas, as well as cow's milk sauce.

○ Food recipes offered by non-PD/ND mothers to their children

The main dishes offered to children were *Akassa* (fermented maize dough); *Wake* (cowpea and rice); pounded yam; spaghetti; *Wassawassa*, a couscous made from yam cossette flour; and *Dambu* (*Peulh*) or *sotè* (*Baatonu*), seasoned maize or sorghum grits. For side dishes, children consume tomato and okra leaves sauce.

C. CESSATION OF BREASTFEEDING

With the exception of *Fulanis*, the age at which PD mothers stop breastfeeding their children is slightly higher than that of non-PD/ND mothers. Indeed, they breastfed their children for a maximum of three years. In addition, PD mothers have a large knowledge of methods for the cessation of breastfeeding. Among the *Gandos*, a PD mother used a so-called “gentle” method to facilitate the cessation of breastfeeding. Instead of abrupt cessation of breastfeeding, they gave to child certain foods to help them forget about breast milk. For example, a *Gando* PD mother used braised cow's milk cheese, to which breast milk is added, to feed their children. In general, all mothers continued to breastfeed their children between the ages of one and two years.

Care practices

Children of PD mothers received more affection than other children (Table 3). The parents of these

Table 3. Care practices of different categories of mothers in the three ethnic groups.

Categories	Mothers' practices in child nutrition		
	Common	Positive deviant	Non-positive deviant/negative deviant
Care	Child is left at home when he starts crawling^G	- The mother took the child with her when she goes out, even when the child begins to walk^B - The child is left at home when he begins to walk^F	- The mother took the child with her when she goes out, until he starts walking^B - The child is left at home when he starts crawling^F
	The child is entrusted to an experienced caregiver in the absence of the mother^{BGF}	No specific practices^{BGF}	No specific practices^{BGF}
	No common practices	The mother spent about 30 minutes playing with the child^{BGF}	The mother did not spend time playing with the child^{BG}
	Child sleeps under a mosquito net^{BGF}	No specific practices^{BGF}	No specific practices^{BGF}
	No common practices^{BGF}	Good supervision of children by parents^{BGF}	Parents supervised their children less^{BGF}
	The children were often reprimanded and beaten when they misbehave^{BG}	The children were reprimanded or beaten, no more than once a week, when they misbehave^F	The children were reprimanded or beaten 4 times a week when they misbehave^F
	No common practices^{BGF}	Mother is very affectionate towards her child^{BGF}	Mother is less affectionate towards her child^{BGF}

^B= Baatonu, ^G= Gando, ^F= Fulani, Several exponents represent several ethnic groups,

children, especially the mother, spend at least 30 minutes playing with their child. Among PD *Baatonu* and *Fulani* mothers, the children were under their mother's care and followed her wherever she moved until they were able to walk. Among non-PD/ND mothers, when the child begins to crawl, he is left at home with collateral relatives who provide secondary care. Among *the Fulani* in particular, PD mothers reprimanded or beat their children less often than other mothers.

Hygiene practices

Hygiene practices were addressed from several aspects (Table 4): food, personal and environmental hygiene. Personal hygiene was studied through the child's bathing and the mother's hand washing. Among the *Baatonu* and *Gando* ethnic groups, PD mothers reported bathing their children up to three times a day. In addition, they wash their hands at three critical times (after handling a child's feces or cleaning a child's buttocks, before preparing food, and before eating the

food), while non-PD/ND mothers wash their hands only at two critical times (before preparing food and before eating the food). In addition, PD mothers of all ethnic groups store their drinking water in covered containers and keep their yards and kitchens clean.

Health practices

Table 5 shows the different maternal health practices. In terms of family planning, birth control was not perceived in the same way by mothers of all ethnic groups. While *Baatonu* PD mothers wanted to have as many children as possible, *Fulani* PD mothers wanted to practice family planning.

Treatment of diarrhea was done only by self-medication with herbal teas or medicines among non-PD/ND *Baatonu* and *Gando* mothers, while PD mothers, even if they initially attempted self-medication, ultimately took their child to a health facility for appropriate care. In these same ethnic groups, certain foods, such as solid foods (cowpeas, maize), were avoided in the child's diet when he had diarrhea.

Table 4. Hygiene practices of different categories of mothers in the three ethnic groups.

Categories	Mothers' practices in child nutrition		
	Common	Positive deviant	Non-positive deviant/negative deviant
Hygiene	• Bathing the child twice a day ^F	• Bathing the child 2 to 3 times a day ^{BG}	• Bathing the child twice a day ^{BG}
	• No common practices ^{BGF}	• Mother wash their hands at 3 critical moments ^{BGF}	• Mother wash their hands at 2 critical moments ^{BGF}
	• Uncovered drinking water jar ^F	• Covered drinking water jar ^{BG}	• Uncovered drinking water jar ^{BG}
	• No common practices ^{BGF}	• Courtyard and clean kitchen area ^{BGF}	• Courtyard and dirty cooking space ^{BGF}

^B= Baatonu, ^G= Gando, ^F= Fulani, Several exponents represent several ethnic groups

Table 5. Health practices of different categories of mothers in the three ethnic groups

Categories	Mothers' practices in child nutrition		
	Common	Positive deviant	Non-positive deviant/negative deviant
Health Family planning	• The mother wants to have as many children as possible^G • The mother is unfamiliar with the practice of birth spacing^{BGF}	• Mothers want to have as many children as possible^B • Mothers want to have a limited (5 to 7) number of children^F	• Wish to have between 4 and 6 children^B • Mothers want to have as many children as possible^F
Treatment of the disease	• Do not avoid any particular food^B	• Avoid solid foods (cowpeas, rice, and maize) in case of diarrhea^{GF}	• Do not avoid any particular food^{GF}
	• Practice of self-medication but go to hospital afterwards^B	• Practice of self-medication but go to the hospital afterwards^{GF}	• Practices only self-medication^{GF}
	• Does not use ORS solution for the treatment of diarrhea^{BGF}	• No specific practices^{BGF}	• No specific practices^{BGF}
	• Recognize signs of illness among children (diarrhea and fever)^{BGF}	• No specific practices^{BGF}	• No specific practices^{BGF}
	• Consult the mother-in-law or husband when the child is sick^{BGF}	• No specific practices^{BGF}	• No specific practices^{BGF}
	• Vaccinated Child^{BG} • Unvaccinated child^F	• No specific practices^{BGF}	• No specific practices^{BGF}

^B= Baatonu, ^G= Gando, ^F= Fulani, Several exponents represent several ethnic groups

Discussion

Feeding practices

One of the main reasons given by mothers for their difficulty in practicing exclusively breastfeeding is the lack of breast milk (20). To address these problems,

PD Gando mothers consumed a porridge with a decoction of *Euphorbia balsamifera*. The leaves of *Euphorbia balsamifera* used by PD mothers are known for their galactogenic effects (21).

The average age (8±1.7 and 6±0 months) of complementary food introduction among PD mothers met the WHO's recommended age, which stipulates

a rigorous practice of exclusive breastfeeding for up to six months (19). However, all *Baatonu* PD mothers gave herbal teas to their children before the recommended age of six months. According to Victora (22), one of the main reasons for discouraging the introduction of herbal teas to children before the recommended age is the association between this practice and the incidence of diarrhea. This link is likely due to poor hygiene during the preparation and administration of these teas. Given that the results of this study show that PD mothers follow good hygiene practices regarding handwashing and drinking potable water, it could be assumed that those who give herbal teas to their children before the age of six months would also follow good hygiene practices when preparing and administering these herbal teas. This is beneficial for their children and has a positive impact on their health. In addition, the drinking water used by these mothers is usually clean and stored in covered containers, which could limit contamination.

In terms of complementary foods, PD mothers gave to their children porridge enriched with legumes or prepared special sorghum porridge (light porridge) for children with cow's milk. Non-PD/ND mothers feed their children with family porridge, which is often very consistent. The legumes and cow's milk incorporated into the porridge provide protein, which is essential for child growth (23,24). Contrary to consistent porridge, the light porridge prepared by PD mothers is more suitable for young children, as it allows them to consume larger quantities (25).

Furthermore, dietary diversity score, the quantity of food served to children, and the daily frequency of consumption score were higher among PD mothers compared with others. PD mothers practice active feeding (accompanied by songs, affection, and encouragement), unlike other mothers. These results were consistent with those of Kanani & Popat (26) and Sethi et al. (27) conducted in India. Active feeding practices are important because they enhance the child's interest in the foods offered and facilitate increased food intake (28).

In addition, children's dietary diversity score (5) and their food consumption frequency score (4) among PD *Gando* and *Fulani* mothers were consistent with the recommendations for children aged 6 to 23

months (19). In their studies on the effect of maternal eating behaviors on the nutritional status of children in Ghana, Nti & Larrey (29) found a significantly high daily food frequency and dietary diversity in positive deviant children. Mackintosh et al. (30) also found that positive deviant mothers fed their young children more times in a day compared to other mothers in Vietnam. This practice would allow children to have the necessary nutrients in quality and quantity for their proper development.

The amount of food served to the child is essential for their nutrition. PD *Baatonu* and *Fulani* mothers served larger quantities of food to their children. The more food served, the more food eaten, which offers to children more chance to cover their nutritional needs (31). Indeed, to enable their children to consume more food, PD *Baatonu* and *Gando* mothers used several methods to stimulate their appetite. These practices could explain the high amount of food served to children of *Baatonu* PD mothers. Appetite plays a key role in the good nutrition of the child since it is a determinant factor for the nutritional status of children (32,33). Furthermore, the low frequency of out-of-household feeding and the limited use of street foods for child feeding enable PD mothers to maintain greater control over their children's diets compared with others.

In terms of the dishes, whether main dishes or complementary recipes, PD food preparations were characterized by a combination of several food groups, like starchy foods (cereals and tubers), legumes, and leafy vegetables. Starchy foods mainly provide energy, while legumes are rich in protein and green leafy vegetables in micronutrients (19,34,35). The presence of green leafy vegetables such as *Senna obtusifolia* and *Urena lobota* leaves provides a significant micronutrient in dishes served to children. Based on the recommended nutritional compositions of fortified complementary foods, 100g of fresh *Senna obtusifolia* leaves provide at least 40% of the recommended content for iron, calcium and vitamin A (35,36).

In general, all mothers continue to breastfeed their children between the ages of one and two years, which is in line with the WHO recommendation (19). Compared with non-PD/ND mothers, the age at which PD mothers stopped breastfeeding was slightly higher

and the cessation was carried out smoothly. Continuing breastfeeding up to two months is very beneficial for the child, especially in cases of illness or loss of appetite, and also helps prevent child death due to infections (37,38). In their study on the association between breastfeeding and the growth of children living in rural areas in Senegal, Simondon et al. (39) reported a significant increase in the height of breastfed children beyond two years of age. The cessation of breastfeeding must be done gently to facilitate the child's transition from a mixed diet to a family diet. This prevents the child from falling into malnutrition, as this period is very critical, since it presents a high risk of malnutrition for children (40).

Care, hygiene and health practices

The mother-child relationship was highly affectionate among PD mothers and their children. Through close supervision, playing, and absence of corporal punishment, PD mothers promoted healthy cognitive development and enabled their children to reach their full potential (41). Talking and playing with the child promotes early stimulation, which plays a decisive role in their mental and social development.

Moreover, personal, environmental (sanitation) and dietary hygiene have a direct impact on the child's nutrition. The good practices adopted by PD mothers such as bathing children, washing hands, keeping the yard and kitchen clean, covering the meals served and drinking water help to avoid contamination as much as possible, thus reducing the incidence of infections (42). Nti & Lartey (29) reported similar results in Ghana. Indeed, they found that mothers of PD children were more likely to wash their hands before feeding their child and keep their environment clean than others. All of these practices observed among PD mothers could help reduce chronic malnutrition among children through the implementation of good Water, Sanitation and Hygiene (WASH) practices. In fact, good WASH practices contribute to a significant reduction in chronic malnutrition among children (43).

With regard to family planning, this practice was limited. The practice of birth limitation and spacing varies across different categories of mothers and between ethnic groups. The birth limitation desired by

Fulani mothers could be a step towards reducing malnutrition given the strong link between household size and malnutrition (44).

In terms of health, self-medication was generally common among mothers across all ethnic groups. For the treatment of diarrhea, they used herbal teas or medicines. When diarrhea persists, non-PD/ND mothers continued self-medication, whereas PD mothers took their children to a health facility for treatment. In addition, they avoid feeding the baby with solid foods, which is consistent with WHO's recommendations (45). Self-medication carries a high risk of overdose, which can have a long-term impact on the child's health (46). At the same time, they are local and affordable solutions for mothers. It is essential that studies should be conducted on these herbal teas in order to provide mothers with instructions for their use to protect children from the risks of overdose and poisoning (47).

Study limitations

This study was conducted during a period of food abundance, which does not take into account the seasonal variability of food resource availability throughout the year. Indeed, this annual variability, characterized by periods of food scarcity and abundance, could lead to identifying other interesting local resources used by PD mothers during the food scarcity season.

Conclusion

This study reveals the positive deviant practices of *Baatonus*, *Gandos* and *Fulanis* mothers. The PD practices identified were feeding, care, hygiene and health. Feeding practices like the age of complementary food introduction and the frequency of food consumption were higher compared to those of other mothers. PD mothers use lactation-enhancing methods and the "Gentle" breastfeeding cessation method to ensure adequate child feeding. The following PD foods, based on cereals, legumes, green leafy vegetables and milk, *Idouare*, *Dambu*, *Dibou kpé sari*, *Kobsa* and cow's milk sauce have been identified. Regarding care practices, a more

affectionate mother-child relationship characterized PD mothers. For hygiene practices, PD mothers carefully cover their food and drinking water. In addition, they keep the yard and kitchen of their home clean. With regard to health practices, PD mothers take their children to a health facility after self-medication when they have diarrhea. All these practices, already present within the community in terms of public health, constitute a real lever for improving children's nutritional status. Learning by sharing among mothers is crucial for disseminating best practices and encouraging their adoption in order to scale them up. These PD dietary practices should be promoted through nutritional education at the community level for behavior changing to combat malnutrition. Incorporating PD child feeding and nutrition practices into nutrition training programs of community actors like community health volunteers, health agents, NGO agents, etc., could enhance the effectiveness of their interventions.

Ethical authorization

The National Ethics Committee for Health Research of Benin (CNER), through decision No. 19/MS/DC/SGM/DRFMT/CNERS/SA dated 12/02/2021, granted ethical approval No. 43 on 22/12/2020 for the implementation of the NaviNut research project under which this study was conducted. Before the consumption of the meal, the objectives, methodology, and significance of the study were explained in the local language to the mothers, and informed consent was obtained from all participating mothers, verbally and by signature (fingerprint) on the consent form.

Acknowledgements: We would like to thank the donor, the Federal Office for Agriculture and Food, for funding this work through the NaviNut project, to which we also express our gratitude. We also thank the mothers, women's group leaders and field assistants in the villages of *Guinrou*, *Alafiarou* and *Nimbéré-Peulh* for their collaboration in the implementation of the activities of this study. A thank you to Mr. JIMMY Kouété for his advice in the writing of this article.

Conflict of Interest: Each author declares that he or she has no commercial associations (e.g. consultancies, stock ownership, equity interest, patent/licensing arrangements etc.)

that might pose a conflict of interest in connection with the submitted article.

Authors Contribution: All authors have accepted responsibility for the entire content of this manuscript and approved its submission.

Declaration on the Use of AI: None.

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